

ScaleOn Study Portal

Full-Stack Internship Management and Learning Portal

Summer Internship Project Presentation

Based on the submitted project report

Project Overview

A full-stack web application for managing internship programs and learning activities.

Centralizes intern accounts, batches, roles, learning modules, assignments and progress.

Also supports sessions, attendance, notifications, support tickets and audit logging.

The documented architecture uses separate frontend and backend applications.

Problem Statement & Objectives

Problem: Internship programs require coordination of users, batches, learning, assignments, communication and administration.

Objective: Build a centralized, maintainable platform for internship lifecycle and learning management.

Provide secure authentication and database-driven role-based access control.

Track learning progress, assignments, sessions and important administrative actions.

Technology Stack

Frontend: Next.js 16, React 19, TypeScript, Tailwind CSS 4

Backend: Node.js 22, Express.js, TypeScript

Database: PostgreSQL

ORM: Prisma

Authentication: JWT access + refresh tokens with HTTP-only cookies

Security & validation: bcrypt, Helmet, HPP, rate limiting, CSRF protection and Zod

System Architecture

User → Next.js Frontend

Frontend → REST API / Authentication Layer

Express middleware → Authentication, authorization, validation and security checks

Application services/modules → Business logic

Prisma ORM → PostgreSQL database

The architecture separates frontend, backend and database responsibilities.

Major Functional Modules

Authentication: login, logout, password reset, token refresh and session controls

Intern Management: accounts, status, batches, roles and transfers

Learning: phases, modules, progress, XP and streaks

Assignments: creation, submission, review, scoring and feedback

Sessions & Support: live sessions, attendance and support tickets

Notifications and Audit: system notifications, activity logs and administrative audit logs

Security & Role-Based Access Control

Short-lived JWT access tokens and rotating refresh tokens

HTTP-only cookies and hashed refresh tokens

bcrypt password hashing and login rate limiting

CSRF protection, Helmet security headers and HPP

Zod request validation and Prisma parameterized queries

Documented roles: Super Admin, Admin, Mentor and Intern

Database & API Design

PostgreSQL database documented with 29 tables.

Main areas: identity, authorization, internship, security/audit, learning, assignments, sessions, support and notifications.

REST-style API using GET, POST, PUT, PATCH and DELETE.

Documented API modules include /auth, /users, /admins, /interns, /roles, /batches, /sessions and /profile.

Assignment & Learning Workflow

Administrators can create, edit and delete assignments.

Assignments can be filtered by learning module and due-date status.

Intern submissions can be reviewed with scores, feedback and status decisions.

Learning modules record completion, score, time, XP and streak information.

Project Setup & Execution

Clone the project repository.

Install backend dependencies inside /backend.

Install frontend dependencies inside /frontend.

Configure backend environment variables and security secrets.

Generate the Prisma client and run backend and frontend development servers.

Open the admin or intern login route in a browser.

Limitations & Future Enhancements

Production deployment strategy is documented as future/TBD.

Observability, health checks, metrics and APM are future areas.

File storage, email infrastructure and real-time notifications are future decisions.

Redis caching, advanced search and expanded mentor workflows are planned.

Future work also includes richer analytics and automated CI/CD pipelines.

Conclusion

ScaleOn Study Portal provides a structured foundation for internship administration and learning.

It combines a modern frontend, TypeScript/Express backend and PostgreSQL/Prisma data layer.

The documented system covers intern lifecycle management, RBAC, learning, assignments, sessions, support, notifications and auditability.

The report identifies additional infrastructure and operational work needed for production-scale deployment.

References

ScaleOn Study Portal GitHub repository

Repository README.md

ARCHITECTURE.md

DATABASE.md

API_DOCUMENTATION.md